CUE-CONDITIONED RECONSTRUCTION: LEARNED DECOMPRESSION FOR CAPACITY-BOUNDED AGENT MEMORY

Anonymous Authors[1]

ABSTRACT

An agent that must hold its memory within a fixed budget has to destroy information, and the question is which representation of the survivors best supports later recall. Current systems compress heuristically – truncate, summarise, or delete – with no model of what a query will need. We formulate memory compaction as *lossy source coding with side information at the decoder*: the encoder compacts a trace without knowing the future query, while the decoder sees contextual cues the encoder did not. This is the Wyner-Ziv setting, and it implies a compaction rule that differs sharply from summarisation: discard preferentially what the cues will make recoverable, and spend bits on what they will not.

We instantiate this as a trained encoder-decoder pair over an agent memory store, with a decoder that emits a calibrated confidence and abstains rather than reconstruct when the cues are insufficient. Training requires no annotation: every compaction produces a (compacted trace, cues, original) triple automatically, so the compressor generates its own supervision.

We first measure the ceiling. On a 1,115-memory corpus compacted to gist, the answer to a question survives compaction for 46% of questions, is absent but latent in neighbouring memories for 7%, and is unrecoverable for 48%. A learned decoder can therefore address at most the middle band, and any system that produces fluent output on the remaining 48% is confabulating by construction. We argue this decomposition, not reconstruction accuracy, is the correct way to evaluate learned memory decompression, and we report abstention quality as the primary metric.

Table 1. Where the answer goes when a memory is compacted to gist. Measured on 147 evidence-annotated questions over a 1,115-memory store.

After compaction	Share	Implication
answer survives in the gist	46%	decoder not needed
lost, but latent in neighbours	7%	**addressable**
unrecoverable	48%	must abstain

1 INTRODUCTION

1.1 Contributions

2 MEMORY COMPACTION AS WYNER-ZIV CODING

3 THE RECOVERABILITY CEILING

4 METHOD

4.1 Encoder: what to discard

4.2 Decoder: reconstruct or abstain

4.3 Escalation

4.4 Objective

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_1 \mathcal{L}_{\text{cal}} + \lambda_2 \mathcal{L}_{\text{abstain}} + \lambda_3 \mathcal{L}_{\text{rate}} \tag{1}$$

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the Machine Learning and Systems (MLSys) Conference. Do not distribute.

5 TRAINING DATA

6 EVALUATION PROTOCOL

7 RELATED WORK

8 CONCLUSION

REFERENCES